Diophantine consistency equations with powered saturation and factorial carriers

Round 14 research report and reproducible artifacts

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Abstract

This iteration improves two explicitly calibrated approaches to an integer polynomial whose unsolvability is equivalent in ZFC to Con(ZFC). The principal small-variable equation has twelve unrestricted integer unknowns, exact degree 768, and 141 live binary arithmetic operations. Its new ingredients are fifth-power divisibility saturation, full supported-modulus Pell interpolation, and cancellation-aware integral centering of a four-root norm. A separate direct proof-certificate equation has 519 unrestricted integer unknowns, degree four, 6253 collected monomials, and coefficient height 411060. A shared-factorial architecture yields a quartic in 349 nonnegative integers; a signed companion has 517 integer variables at a larger height. These are measured tradeoffs, not global optimality claims. All new unbounded arguments are proved below; inherited native and logical calibration boundaries are identified rather than presented as newly kernel-checked theorems. Exact equations, actual coefficient payloads, source pins, generators, independent checkers, and clean-replay instructions accompany this report.

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1 Results and precise interpretation

1.1 The selected equations

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$; "positive" means strictly greater than zero. A named arithmetic register is an abbreviation, not an additional quantified variable. Coefficient height is the maximum absolute value of a fully expanded polynomial's integer coefficients. Arithmetic operations count live binary addition, subtraction, and multiplication, including squares and multiplication by a fixed scalar; register reuse and fixed leaves cost zero.

Endpoint and domain	Variables	Degree	Rows	Terms	Height
Centered norm, $\mathbb{Z}$	12	768	—	not expanded	not measured
Direct centered, $\mathbb{Z}$	519	4	321	6253	411060
Derived centered, $\mathbb{Z}$	517	4	310	6774	3904267
Derived labels, $\mathbb{N}$	349	4	268	12768	86915503

The norm uses 141 operations: 88 multiplications, 26 additions, and 27 subtractions. Its 149-operation reduced graph is the same polynomial, not merely an equivalent zero-existence condition. The norm keeps all six authentic fixed coefficients from the pinned source and is not a small-literal-coefficient construction. By contrast every coefficient of the direct 519-variable quartic has absolute value at most $411060 < 2^{19}$. Its residual count is not an operation count.

For each endpoint P over its stated domain D, the intended statement is

$$\mathrm{ZFC} \vdash \left[\forall x \in D^n \; P(x) \neq 0\right] \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}). \tag{1}$$

The ordinary mathematical proof factors through

$$(\exists x \in D^n \; P(x) = 0) \quad \longleftrightarrow \quad \text{there is a finite syntactic ZFC refutation.}$$

This is an equivalence, not a proof of consistency. No consistency assumption is used. This report supplies ordinary proofs and finite exact algebraic verification, not a newly completed Lean or Rocq development or a separately encoded object-language derivation in ZFC. The inherited formalizability and native-calibration boundary remains explicit.

1.2 What genuinely changed

The norm replaces full carrier divisibility by fifth-power divisibility. A quantitative separation bound from the authentic large base forces the same exact main projection. A stronger p-adic interpolation theorem removes a redundant positivity-witness factor from the strong Pell scale. Finally, an integral shear cancels the highest norm terms.

The direct branch uses reciprocal prime-power CRT with a small self-bounding factorial, a carrier-derived amplification exponent, an exact signed factorization of the Horn certificate, and certified integral projections and coordinate translations. The natural branch uses the same factorial both to separate formula labels and to supply CRT moduli; a derived polynomial range eliminates an extra local slack. This direct route uses neither a Jones universal equation nor K215. The norm retains the source34 tuple and its own inherited computational provenance; these routes must not be conflated.

1.3 Pinned sources and proof boundaries

The public comparison snapshot is ProveIt commit

`84e16e43092614bae262ca59651cb9d77e31a255`.

The supplied prime-power and reciprocal-shell packets retain logical calibration at commit `08200eb10fde2db408c69dffd840d6920a569b1f`; some direct developments are ahead of public main. The norm keeps the source34 six-port tuple, whose binding was recorded at `2605669a532342f153b13eb6b4a0af4b54f5fa37`. Source pins and hashes are included.

For the direct branch, the named calculus boundary is `KEndpoint.c222_iff_zf_refutation`, followed by `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. Source-rule syntax, the original Horn interface, simultaneous Pell powers, binomial digit graphs, and nonnegative contraction are retained from supplied proofs. For the norm, the inherited interfaces are scalar-parent radical separation, adjacent-coefficient descent, signed folding, pre-projection ranks, canonical witnesses, and source34 calibration. Their proof texts are bundled. The new saturation argument does not assume a main projection before recovering it.

The known shared83 and weakened86 counterexamples concern different weakened arithmetic interfaces, not the retained source34 calibration. The former loses required carrier divisibility; the latter loses a strict positivity slack. Neither refutes the modified75 compiler itself. Here the full signed kernel is restored before invoking its inherited ordinary calibration, and the strict determinant guard remains. The targeted source review is included in the archive.

2 The twelve integer equation

2.1 A complete polynomial definition

The independent coordinates are

$$j, C, z, h, r, v, w, T, c, n, t, s \in \mathbb{Z}.$$

The six fixed ports

$$\beta, K_0, M_C, M_F, A_{\mathrm{cone}}, B_{\mathrm{cone}}$$

are the complete unchanged source34 tuple, including the already shifted M_F. In particular $\beta = 2^{1064921} - 1$. All six full values are supplied as compressed minimal unsigned little-endian bytes, not placeholders. Define

$$\begin{aligned}
J &= 6j + 1, \qquad q = \beta J + 1, \qquad W_0 = 12w + 4,\\
X &= qW_0, \qquad Y = (2v + 1)q^3, \qquad E = XY,\\
a &= E + Y, \qquad H = 4a + 3, \qquad \Delta = a^2 + H = (a + 2)^2 - 1,\\
F &= (K_0 + W_0)C - z(q - 1),\\
R &= [q(q - F) - C](q^2 - 1) + (M_C + qM_F)J,\\
k &= R + 1 + hE, \qquad \eta = c - Yk, \qquad \zeta = k - \eta,\\
U &= A_{\mathrm{cone}}F, \qquad W = B_{\mathrm{cone}}C,\\
G &= UW - (q - U - W)^2,\\
\mathrm{Pair} &= G\eta\zeta - (t^2 - G - \eta\zeta + 1)^2,\\
D_g &= (3\,\mathrm{Pair} - 4)s^2 + 1,\\
I &= rc^2, \qquad D_f = 1 + \Delta I^2, \qquad Q = \Delta^2 I^2,\\
\delta &= Q - 1, \qquad C_y = QV^2 - 1,\\
V &= D_f(c - R + 4IT) - c,\\
\ell &= XY^2, \qquad D_\tau = 1 + \ell(\ell + 1)k^2,\\
K_{\mathrm{red}} &= X(2X - 3c) - 2, \qquad z_c = HT + K_{\mathrm{red}},\\
A_c &= (z_c^2 + 3H^2)(z_c^3 - 19H^3),\\
S &= srH, \qquad Z = S(n + QV) - A_c.
\end{aligned}$$

Define the sign-invariant polynomial

$$\mathcal{F}_{16}(Z;u_1,u_2,u_3,u_4)=\prod_{\varepsilon\in\{\pm1\}^4}\left(Z-\sum_{i=1}^{4}\varepsilon_i\sqrt{u_i}\right).$$

Independent sign invariance makes it an ordinary integer polynomial. The equation is

$$\boxed{P_{12}=\mathcal{F}_{16}(Z;9S^2D_f,81S^2D_\tau,729S^2D_g,S^2\delta C_y)=0.} \tag{2}$$

The literal graph inlines all definitions with only integer leaves and $+,-,\times$; it contains no radicals, divisions, variable powers, or additional unknown registers. The shear $n_{\mathrm{old}}=n+QV$ is a bijection of $\mathbb{Z}$ for fixed other coordinates, because QV does not involve n.

2.2 Guard recovery before the strong scale is nonzero

The unchanged unconditional front end gives q even, $q\equiv 2 \pmod 3$, $|q|\ge\beta-1$, $|X|\ge 4|q|$, $|Y|\ge 4096$, H odd and nonzero, $\Delta>1$, $D_f,D_\tau\ge 1$, and $\delta\neq 0$. Modulo three,

$$H\equiv\Delta\equiv Q\equiv 0,\quad D_f\equiv D_\tau\equiv 1,\quad \delta\equiv C_y\equiv -1.$$

If $S=0$, every radicand vanishes, leaving A_c^{16}. But

$$A_c=H^5\pi\left(T+\frac{K_{\mathrm{red}}}{H}\right),\qquad \pi(x)=(x^2+3)(x^3-19)$$

is nonzero: the quadratic has no real zero, and the rational-root theorem excludes a rational cube root of 19. Thus $s,r\neq 0$. This does not yet prove $c\neq 0$.

At a zero divide a vanishing conjugate factor by S. The inherited prime-order separation lemma applies to the normalized roots. Their squared 3-adic orders, in the order auxiliary, $3\sqrt{D_f}$, $9\sqrt{D_\tau}$, $27\sqrt{D_g}$, are

$$0,\quad 2,\quad 4,\quad 6+v_3(D_g),$$

with a zero guard root omitted. This remains valid at $I=0$, where $\delta=C_y=-1$. The signed rational sum forces each root rational. Thus $D_g=g^2$ for an integer g; since $s\neq 0$, Pair ≤ 0 would make $D_g<0$, so Pair >0.

For integers u,v, if $uv-(t^2-u-v+1)^2>0$, then $uv>0$. Both cannot be negative, because the squared term would be at least $(|u|+|v|+1)^2>uv$. Hence $u,v>0$. Apply this with $u=G$, $v=\eta\zeta$ to get $G>0$ and $\eta\zeta>0$. If $c=0$, however,

$$\eta\zeta=-Y(Y+1)k^2\le 0.$$

Therefore $c\neq 0$, and only now $I=rc^2\neq 0$ and $Q>1$.

The retained adjacent-coefficient Pell descent applied to the extracted rational auxiliary root gives an integer y with

$$QV^2-(Q-1)y^2=1. \tag{3}$$

For the integral radicand, divide the normalized integer root by δ before this descent. The other roots are integers f,τ satisfying $f^2=D_f$ and $\tau^2=D_\tau$. Every sign of I,V,c,a remains allowed. The integral factor gives $S\mid A_c$ and therefore

$$H\mid K_{\mathrm{red}}^5. \tag{4}$$

It would be invalid to infer $H\mid K_{\mathrm{red}}$ from divisibility alone: $243\mid 3^5$ but $243\nmid 3$.

2.3 The inherited preprojection index interface

For any integer A, the Pell polynomials are defined by $\chi_A(0) = 1$, $\chi_A(1) = A$, $\psi_A(0) = 0$, $\psi_A(1) = 1$, and the recurrence $u(n+2) = 2Au(n+1) - u(n)$. Thus the signed-base expressions below are defined even when $a + 2 < 0$.

The scalar-parent signed folding and index proof requires: the unchanged cubic Y and packing definitions; $G > 0$ and $\eta\zeta > 0$; $I \neq 0$, $c^2 \mid I$, $|I| \geq |c|$; the two Pell equations; and

$$V \equiv -c \pmod{f}, \qquad V \equiv -R \pmod{c}.$$

These follow from the definitions and

$$V + R = \Delta I^2(c - R) + 4ID_fT.$$

Neither $s \mid I$ nor an odd strong multiplier is needed.

For the precise chronology, put $B_0 = |a+2|$, $C_0 = |c|$, $I_0 = |I|$. Pell classification gives $|f| = \chi_{B_0}(m_0)$, $I_0 = \psi_{B_0}(m_0)$. The auxiliary Pell index at $\rho = \Delta I_0$ is odd. Reduction modulo $|f|$ and folding modulo $2m_0$ gives $C_0 = \psi_{B_0}(p)$; $2I_0 < |f|$ excludes wrapping. Strong divisibility and $C_0^2 \mid I_0$, through the Lucas multiplication formula, give $p \mid m_0$ and $C_0 \mid m_0$. The auxiliary residue gives $R \equiv \pm p \pmod{C_0}$; the first-index bounds $0 < R, p < C_0/2$ force $p = R$. These steps precede and do not use the main projection or (4).

With $x = |X|$, $m = |Y|$, $\sigma = \operatorname{sign}(c)$, the retained rank and ratio estimates then give

$$\begin{aligned} &R = 2r_0 + 1, \quad r_0 \geq 24, \quad c = \sigma\psi_{a+2}(R), \\ &m > \frac{(x-3)^{r_0}}{2} - 1, \qquad |H| > x(x-3)^{r_0}. \end{aligned} \tag{5}$$

Indeed $|H| \geq 4(x-1)m - 3$, and the remaining difference exceeds $(x-2)(x-3)^{r_0} - 4x + 1 > 0$ for $x \geq 64, r_0 \geq 1$. The complete inherited folding and rank proofs are bundled. This is their exact interface, not a claim that finite arithmetic tests prove it.

2.4 Powered divisibility restores the exact projection

Theorem 2.1 (Fifth-power saturation)**.** *Every integer zero of* (2) *restores the same exact signed main projection as the pinned scalar source.*

Proof. Set $K = X(X + 2ac) - 1$ and $\Pi = (\sigma X - 2^R)(2^R\sigma X + 1)$. The exact identity and retained signed Pell congruence are

$$2K = K_{\mathrm{red}} + HcX, \qquad 2^R K \equiv \Pi \pmod{H}. \tag{6}$$

Since H is odd, (4) gives $H \mid K^5$. Taking fifth powers in the congruence gives $H \mid \Pi^5$. No nonunit is cancelled modulo a composite H.

For every raw integer assignment, the authentic tuple gives $x \geq 4(\beta - 1) \geq 2^{64}$. Both factors of Π have absolute value less than $2^{R+1}x$, so

$$|\Pi|^5 < 2^{10R+10}x^{10} = 2^{20r_0+20}x^{10}.$$

By (5) and $x - 3 > x/2$, $|H| > x^{r_0+1}/2^{r_0}$. For $r_0 \geq 24$,

$$64(r_0 - 9) - (21r_0 + 20) = 43r_0 - 596 \geq 436 > 0.$$

Thus $x^{r_0-9} > 2^{21r_0+20}$ and $|\Pi|^5 < |H|$. Divisibility forces $\Pi = 0$. Its second factor is odd and nonzero, so $\sigma X = 2^R$. Put $d = \sigma\chi_{a+2}(R)$. The signed projection congruence now gives $H \mid d - X - ac$, restoring

$$\gamma = (d - X - ac)/H \in \mathbb{Z}.$$

All other source rows and guards are unchanged, so the generalized signed native theorem gives the same finite-refutation endpoint. □

Lemma 2.2 (A general saturation bound)**.** *Let b, d be positive integers, r a nonnegative integer, and $x \geq 1$. Suppose $|H| > x^{r+1}/2^r$, $|\Pi| < 2^{4r+4}x^2$, $x \geq 2^b$, and*

$$b(r+1-2d) > (4d+1)r + 4d.$$

Then $H \mid \Pi^d$ implies $\Pi = 0$. For all $r \geq r_{\min}$ it suffices that $b > 4d+1$ and the displayed inequality holds at $r_{\min}$.

Proof. The exponent inequality gives $x^{r+1-2d} > 2^{(4d+1)r+4d}$. Therefore $|\Pi|^d < 2^{4dr+4d}x^{2d} < x^{r+1}/2^r < |H|$. An integer multiple of H in that strict range is zero. The margin is affine in r with positive slope under $b > 4d+1$. □

The choice $(b, d, r_{\min}) = (64, 5, 24)$ has margin 436. This is a sufficient bound, not an optimality statement.

2.5 Full supported-modulus Pell interpolation

For $A > 1$, define $\chi_A(n), \psi_A(n)$ by

$$(A + \sqrt{A^2-1})^n = \chi_A(n) + \psi_A(n)\sqrt{A^2-1}.$$

Theorem 2.3. *Let canonical native data satisfy: $A_0 > 1$ is even, $\Delta = A_0^2 - 1$, $R \equiv 3 \pmod 4$, and $c = \psi_{A_0}(R) > 0$ with $c \equiv R \pmod 4$. Fix $s > 0$. Choose one positive even m_0 such that*

$$I = \psi_{A_0}(m_0), \qquad f = \chi_{A_0}(m_0), \qquad 4sc^2 \mid I.$$

For every positive M whose prime factors divide I, this same fixed pair admits every residue of T modulo M in

$$V = f^2(c - R + 4IT) - c, \qquad QV^2 - (Q-1)y^2 = 1, \quad Q = \Delta^2 I^2,$$

with integers V, y, T.

Proof. The inherited finite matrix-period argument supplies the strong pair. Its index valuation lemma gives $v_p(m_0) \leq v_p(I)$ for $p \mid I$, and $\lambda = 4m_0 f$ is a period modulo f^2. Keep the pair fixed. Set

$$\rho = \Delta I, \quad D = \rho^2 - 1, \quad d = \operatorname{lcm}(\lambda, 4I), \quad V_\nu = \chi_\rho(\nu)/\rho \quad (\nu > 0 \text{ odd}).$$

The quotient is integral. Since $\gcd(f, I) = 1$,

$$v_p(d) = v_p(4I) \quad (p \mid I). \tag{7}$$

The inherited base interpolation supplies $\nu_0 \equiv 3 \pmod 4$ and an integral T_0. For $h \geq 0$ let

$$\nu(h) = \nu_0 + dh, \qquad T(h) = \frac{V_{\nu(h)} + c - f^2(c-R)}{4If^2}.$$

This is integral. The period preserves the numerator modulo f^2; the identity $V_\nu \equiv -\nu \pmod{4\rho^2}$ for $\nu \equiv 3 \pmod 4$ preserves it modulo $4I$. The two moduli are coprime. Each index supplies $y = \psi_\rho(\nu(h))$.

We claim the stronger isometry

$$v_p(T(h_1) - T(h_2)) = v_p(h_1 - h_2) \quad (p \mid I,\ h_1 \neq h_2). \tag{8}$$

For odd $\nu > 0$ and a nonzero increment d_0 divisible by four, Pell addition gives

$$V_{\nu+d_0} - V_\nu = V_\nu(\chi_\rho(d_0) - 1) + \psi_\rho(\nu)D\frac{\psi_\rho(d_0)}{\rho}. \tag{9}$$

Negative increments follow by interchanging indices. For $p \mid \rho$, $\psi_\rho(\nu)$ and D are units. The lifting facts are

$$v_p(\psi_\rho(d_0)/\rho) = v_p(d_0), \tag{10}$$
$$v_p(\chi_\rho(d_0) - 1) = 2(v_p(\rho) + v_p(d_0)) \qquad (p \text{ odd}), \tag{11}$$
$$v_2(\chi_\rho(d_0) - 1) = 2(v_2(\rho) + v_2(d_0)) - 1. \tag{12}$$

To prove them, put $U = \chi_\rho(4)$, $B = \psi_\rho(4)$. Explicitly

$$B = 4\rho(2\rho^2 - 1), \qquad U - 1 = 8\rho^2(\rho^2 - 1).$$

For odd $p \mid \rho$, write $b = v_p(B) \geq 1$. In the second coordinate of $(U + B\sqrt{D})^k$, if $p \nmid k$ the linear term $kU^{k-1}B$ has valuation b, and all higher odd terms have valuation at least $3b > b$. If $k = p$, the linear term has valuation $b + 1$, middle odd terms at least $3b + 1$, and the last term valuation $pb > b + 1$, including $p = 3, b = 1$. Iterating the prime-to-p and p steps proves (10) for odd primes. At $p = 2$, ρ is even and U odd; odd multiplication preserves valuation by the same expansion. Every doubling adds exactly one via $\psi_\rho(2n) = 2\chi_\rho(n)\psi_\rho(n)$ with odd $\chi_\rho(n)$. Starting at index four gives the asserted 2-adic formula.

The norm identity

$$(\chi_\rho(d_0) - 1)(\chi_\rho(d_0) + 1) = D\psi_\rho(d_0)^2$$

gives the first-coordinate valuations: the second factor is a unit for odd p, while at two it has valuation one because $\chi_\rho(d_0) \equiv 1 \pmod 8$. Thus in (9) the second term has valuation exactly $v_p(d_0)$ and the first strictly greater. Substitute $d_0 = d(h_1 - h_2)$, divide by $4If^2$, and use (7) to prove (8).

For every $p^e \mid M$, the isometry makes $h \mapsto T(h)$ well-defined and injective on $\mathbb{Z}/p^e\mathbb{Z}$, hence bijective. CRT then supplies any requested T modulo M, using a nonnegative representative of h. This does not extrapolate a low-precision linear congruence to arbitrarily high prime powers. □

2.6 Completeness and quantifier order

Start with the canonical signed native witness of a finite refutation. Keep j, C, z, h, v, w, c. Choose t so Pair > 0, then a positive nonzero s and integer g with $g^2 = (3\,\text{Pair} - 4)s^2 + 1$. The retained positivity theorem supplies this, including Pair $= 1$.

Choose one strong pair with $4sc^2 \mid I$, and set $r = I/c^2$. Then $r > 0$, $4s \mid r$, and every prime factor of $M = sr$ divides I. The native projection gives $H \mid K$ and hence $H \mid K_{\text{red}}$; put $\kappa = K_{\text{red}}/H \in \mathbb{Z}$. The inherited intersectivity theorem for π supplies $a \pmod M$ with $\pi(a+\kappa) \equiv 0 \pmod M$. Theorem 2.3 supplies $T \equiv a \pmod M$ with the same I, f. Hence

$$A_c/S = H^4\pi(T + \kappa)/(sr) \in \mathbb{Z}.$$

Choose

$$n = A_c/S + \delta y + 3f + 9\tau + 27g - QV.$$

A conjugate factor vanishes. The order is native witness, t, s, one fixed strong pair I, f, r, one intersective residue, T, y, and then n. No strong pair changes after choosing a residue. This completes both directions.

A conservative companion keeps $I = src^2$ and $S = srH$, with degree 800 and 141 operations. There $M = sr = I/c^2$ divides I, so the original interpolation theorem suffices. The selected release is the stronger proved degree-768 equation.

2.7 Universal norm identity and exact degree

Put $B = Sn - A_c$, $L = SQV$, $D = S^2[Q(V^2+1)-1]$, and $\alpha = 9S^2D_f$, $\beta_1 = 81S^2D_\tau$, $\gamma = 729S^2D_g$. Then $S^2\delta C_y = L^2 - D$. Pairing auxiliary signs first gives

$$P_{12} = \prod_U ((B-U)^2 + 2L(B-U) + D), \tag{13}$$

where U runs over the eight signed sums of square roots of α, β_1, γ. This polynomial identity holds without positive radicands. The reduced graph makes cancellation explicit at its first quadratic register:

$$q_0 = B^2 + 2LB + D + \gamma - \alpha - \beta_1.$$

The degrees are

register	I	S	Q	V	A_c	B	L	D	α, β_1, γ
degree	3	8	30	23	35	35	61	92	34, 50, 62.

Each factor in (13) has unique top contribution $2LB$, of degree 96. All competitors have degree at most 92, apart from B^2 of degree 70. Thus $(P_{12})_{\mathrm{top}} = (2L_{\mathrm{top}}B_{\mathrm{top}})^8$. The reduced graph also has syntactic degree bound 768, with register bounds $96, 192, 384, 349$ at q_0, v_0, n_0, n_1.

To prove attainment at the actual tuple, put

$$\begin{aligned} q_0' &= 6\beta j, & W_{00} &= 12w, \\ a_0 &= 2v(q_0')^4 W_{00} = 2^7 3^5 \beta^4 j^4 wv, & & \\ F_0 &= W_{00}C - q_0' z, & R_0 &= -(q_0')^3 F_0, \\ B_{\mathrm{top}} &= -4^5 a_0^5 T^5, & L_{\mathrm{top}} &= -4sr^5 c^8 a_0^7 R_0. \end{aligned}$$

Then $(P_{12})_{\mathrm{top}} = 2^{104} s^8 r^{40} c^{64} a_0^{96} R_0^8 T^{40}$, and

$$\begin{aligned} [j^{408} w^{104} v^{96} C^8 s^8 r^{40} c^{64} T^{40}] P_{12} &= 2^{816} 3^{512} \beta^{408} > 0, \\ [j^{416} w^{96} v^{96} z^8 s^8 r^{40} c^{64} T^{40}] P_{12} &= 2^{808} 3^{512} \beta^{416} > 0. \end{aligned}$$

Both exponent sums are 768; none of the other five ports can cancel these terms. The 141-operation graph has naive degree bound 976, not its exact degree. Its reduced graph represents exactly the same polynomial at 149 operations.

The independent auditor constructs the generic norm as the resultant of

$$u^4 - 2(\alpha + \beta_1)u^2 + (\alpha - \beta_1)^2$$

and

$$(Z-u)^4 - 2(\gamma + d)(Z-u)^2 + (\gamma - d)^2,$$

checks all 495 terms against the norm tail, and checks the transformed prefix separately. These exact identities establish transcription and degree facts, not the inherited native calibration.

3 One factorial for proof labels and CRT rows

This section proves the new arithmetic route completely, including its constructive reverse direction. Its final Cantor-syntax endpoint is a quartic in 349 nonnegative integer unknowns. It is the sum of squares of 268 explicitly listed quadratic residuals, or equivalently the supplied polynomial with 12 768 collected monomials and coefficient height 86 915 503. The square-shell companion has 351 natural unknowns, 270 residuals, 16 193 monomials and height 15 317 516.

The inherited interface is the literal C222 rule calculus, its existing ordinary ZFC calibration, and the simultaneous positive-power Pell representation. Those results are not reproved or advertised as newly formalized here. Everything replacing the old large carrier and CRT architecture is

proved below. In particular, no incomplete global encoding is labelled a consistency polynomial. Neither consistency nor inconsistency is assumed, and no new end-to-end Lean or object-language ZFC derivation is claimed.

The structural idea is to reuse $F = t!$ for two distinct purposes: the weights $1 + jF$ separate formula-label multiplicities, while the moduli $1 + F^2/i$ separate the bounded arithmetic rows. A binomial range condition applied to a *derived polynomial expression*, rather than another independently encoded local coordinate, connects the two roles. This removes the former enormous carrier power and its fixed chain of twenty-two squarings.

3.1 The precise local source interface

Natural numbers include zero throughout. Fix a positive horizon t, outer parameters x, y, F, a row index $0 \le i < t$, and eight local coordinates

$$(a, b, c, d, S, S', u, v).$$

Their bounds are

$$a, b, c, d \le t, \qquad S, S' \le y, \qquad u, v \le x. \tag{A1}$$

Let $c_1, \dots, c_9$ be the nine literal axiom/schema codes in the inherited calibrated source, and let I, A be its implication and universal constructors. Both syntax variants use their complete constructor recipes, not unspecified parameters. All constructor expressions are polynomials with nonnegative integer coefficients. Define

$$W_z = 1 + Fz, \qquad G = \big((c-a)^2(c-b)^2 - (d+1)^2\big)^2.$$

The nonnegative Horn factor is

$$\begin{aligned} H = {} & (S'-S)^2 \prod_{\nu=1}^{8} (S'W_{c_\nu} - S)^2 \\ & \cdot \big((S'W_{c_9} - S)^2 + G\big) \\ & \cdot (S'W_b - SW_aW_{I(a,b)})^2 (S'W_{A(a,b)} - SW_b)^2. \end{aligned}$$

The local polynomial is

$$\begin{aligned} L(i, a, b, c, d, S, S', u, v) = {} & H + \big(x - S - (1 + (i+1)y)u\big)^2 \\ & + \big(x - S' - (1 + (i+2)y)v\big)^2. \end{aligned} \tag{A2}$$

It is nonnegative at every natural tuple, not only at bounded ones. The extra expression controlling all possible rule labels is

$$J(a, b, c, d) = \sum_{\nu=1}^{9} c_\nu + a + b + I(a, b) + A(a, b) + 1. \tag{A3}$$

In particular $J \ge 1$. If $J \le t$, every label that can occur in a vanishing Horn branch lies in $[0, t]$.

The finite majorant calculation is performed on these actual source expressions. Put

$$R = 2 + t + x + y + F.$$

Replace every difference square $(r - s)^2$ in the source by $(U + V)^2$, using positive upper bounds for r, s, and bound each local input by its corresponding quantity in (A1). Include both L and J, together with $t + x + y + 1$, in the final upper expression. The complete gate ledger gives

$$0 \le L(i, \mathbf{w}) < R^E, \qquad 0 < J(\mathbf{w}) < R^E, \qquad E = 8\,388\,608. \tag{A4}$$

A separate positive-constructor ledger gives

$$J(a,b,c,d) \le (n+1)^{\ell} \quad (0 \le a,b,c,d \le n,\ n \ge 1), \qquad \ell = 4\,194\,304. \tag{A5}$$

These are exact finite certificates. A constant $c > 0$ is assigned exponent $\lceil \log_2 c \rceil$; additions increase the larger exponent by one; products add exponents. Since each relevant bounding base is at least 2, every such step is an elementary valid inequality. The input bounds and all constructor constants occur in the ledgers. Thus (A4)–(A5) do not conceal coefficient factors in an asymptotic estimate. Their finite values and the full ledger identities are independently checked.

3.2 Reciprocal moduli and the range lemma

Reciprocal-modulus lemma. For $t \ge 1$, set

$$F = t!, \qquad T = F^2, \qquad d_i = 1 + \frac{T}{i} \quad (1 \le i \le t), \qquad B = \binom{T+t}{t}. \tag{A6}$$

Then every d_i is an integer greater than one, its prime divisors exceed t, the d_i are pairwise coprime, and

$$B = \prod_{i=1}^{t} d_i. \tag{A7}$$

Proof. Because $i \mid F$, one has $d_i = 1 + F(F/i) \equiv 1 \pmod{F}$. A prime at most t divides F, and therefore cannot divide d_i. If a prime divided both d_i, d_j, it would divide $id_i - jd_j = i - j$; for distinct i, j, this is impossible because that prime exceeds t. Finally,

$$\prod_{i=1}^{t} d_i = \prod_{i=1}^{t} \frac{T+i}{i} = \frac{(T+t)!}{T!\,t!} = \binom{T+t}{t}.$$

The final arithmetic graph materializes only the binomial on the right. No variable-length product or division by a variable index appears in the polynomial. In particular, $B \ge 2$. □

One-sided prime-power range lemma. Let p be prime, $e, k \ge 1$, and $a \ge 0$. With the usual convention $\binom{a}{k} = 0$ for $a < k$,

$$p^e \mid \binom{a}{k} \quad \Longrightarrow \quad a \bmod p^e < k. \tag{A8}$$

Proof. The case $a < k$ is immediate. Otherwise the factorial valuation formula is

$$v_p\binom{a}{k} = \sum_{j \ge 1} \left(\left\lfloor \frac{a}{p^j} \right\rfloor - \left\lfloor \frac{a-k}{p^j} \right\rfloor - \left\lfloor \frac{k}{p^j} \right\rfloor \right). \tag{A9}$$

Every summand is 0 or 1, by the elementary floor-addition identity. If the interval $[a-k+1, a]$ contained no multiple of p^e, it would contain no multiple of p^j for $j \ge e$. For those j, its length is less than p^j, and the corresponding summands in (A9) vanish. Hence the valuation would be at most $e-1$, a contradiction. Therefore $p^e \mid a - r$ for some $0 \le r < k$. The least residue of a is the least residue of r, which is at most r, proving (A8). □

This implication is deliberately one-sided. For example $4 \bmod 4 = 0 < 2$, but $4 \nmid \binom{4}{2} = 6$. The construction never invokes the false converse.

3.3 Complete amplified bounded-universal elimination

Here is the general arithmetic interface used below. Let $b_j \geq 0$, let $t \geq 1$, and let

$$K \geq \max(1, t, b_1 + 1, \ldots, b_r + 1).$$

Use (A6) and put $N = B^K$. Let $P(i, \mathbf{w})$ be an integer polynomial, with any fixed external parameters suppressed, such that

$$0 \leq P(i, \mathbf{w}) < 2^K \quad \text{for } 0 \leq i < t,\ 0 \leq w_j \leq b_j. \tag{A10}$$

Then the following statements are equivalent:

1. For every $i < t$, there is a bounded tuple $\mathbf{w}$ with $P(i, \mathbf{w}) = 0$.
2. There are positive integers $C, A_1, \ldots, A_r \leq N^2$ such that

$$B \mid C + T + 1, \tag{A11}$$

$$N \mid \binom{C}{t}, \qquad N \mid \binom{A_j}{b_j + 1} \quad (1 \leq j \leq r), \tag{A12}$$

$$N \mid P(C, \mathbf{A}). \tag{A13}$$

For soundness the upper bounds $C, A_j \leq N^2$ are not needed.

Soundness. Fix $i \in \{1, \ldots, t\}$, choose a prime $p \mid d_i$, and write $m = v_p(d_i) \geq 1$, $e = Km$. Pairwise coprimality gives $v_p(N) = e$. Since $T \equiv -i \pmod{p}$, (A11) implies $C \equiv i - 1 \pmod{p}$. Applying (A8) to the index range gives $C \equiv s \pmod{p^e}$ for some $0 \leq s < t$. As $p > t$, the congruence $s \equiv i - 1 \pmod{p}$ forces $s = i - 1$. For each local column, (A8) similarly gives $A_j \equiv w_j \pmod{p^e}$ with $0 \leq w_j \leq b_j$. Polynomial congruence and (A13) now give

$$p^e \mid P(i - 1, \mathbf{w}).$$

But $p^e \geq 2^K$, so (A10) forces this nonnegative value to be zero. This produces a valid bounded row for each index.

Constructive completeness. Select one bounded tuple for each row. Because the d_i are pairwise coprime, CRT supplies representatives in $[1, B^{2K}]$ satisfying

$$C \equiv i - 1 \pmod{d_i^{2K}}, \qquad A_j \equiv w_{i-1,j} \pmod{d_i^{2K}}. \tag{A14}$$

A zero least residue is replaced by B^{2K}. Thus the representatives are positive and at most N^2. Equations (A11) and (A13) follow immediately.

For any requested range lower index $k \leq K$, a representative a in (A14) is congruent to some $w \in [0, k - 1]$ modulo p^{2Km}. If $a < k$, its binomial is zero. Otherwise the falling factorial has the positive factor $a - w$, divisible by p^{2Km}. The elementary valuation bound

$$v_p(k!) \leq k - 1 \leq K - 1 < Km$$

shows that division by $k!$ still leaves valuation at least Km. This holds for every prime divisor of B, proving all conditions in (A12). The strengthening to precision $2K$ is exactly what makes completeness valid even when a small prime divides a range denominator. □

The free index C and its range condition are essential. One must not reuse the old unamplified reciprocal index $C = B - T - 1$. For $t = 2$, $F = 2$, $T = 4$, $B = 15$, this gives $C = 10$: it is 0 modulo 5, but not modulo 25. The frozen construction retains (A11)–(A12).

3.4 A self-bounding factorial, with both implications

Factorial lemma. Let $n \geq 1$ and let $f \geq 0$ be a proposed factorial output. Put

$$\rho = fn^2, \qquad V = (\rho+1)^n, \qquad D = \binom{\rho+n}{n}.$$

There exist natural u, v with

$$V + D = Df + u + 1, \qquad D = u + v + 1 \tag{A15}$$

if and only if $f = n!$.

Proof of soundness. The second row gives $D \geq 1$, and subtracting it from the first gives $V + v = Df$. Since $V \geq 1$, one first obtains $f \geq 1$. The rows say exactly

$$f - 1 < \frac{V}{D} \leq f, \qquad f = \left\lceil \frac{V}{D} \right\rceil. \tag{A16}$$

Write $G = n!$. The ratio is

$$\frac{V}{D} = G \prod_{i=1}^{n} \frac{\rho+1}{\rho+i} = G \prod_{i=1}^{n} \left(1 - \frac{i-1}{\rho+i}\right). \tag{A17}$$

Every factor is positive and at most 1. Moreover

$$\sum_{i=1}^{n} \frac{i-1}{\rho+i} \leq \frac{n(n-1)}{2\rho} = \frac{n-1}{2fn} < \frac{1}{2f},$$

with the zero sum for $n = 1$ satisfying the same final strict inequality. The elementary finite-product estimate $\prod_i (1 - z_i) \geq 1 - \sum_i z_i$, for $0 \leq z_i \leq 1$, therefore gives

$$G\left(1 - \frac{1}{2f}\right) < \frac{V}{D} \leq G. \tag{A18}$$

Since G is an integer, (A16) and the upper bound give $f \leq G$. The lower bound and $V/D \leq f$ give

$$G < \frac{2f^2}{2f-1} = f + \frac{f}{2f-1} \leq f + 1.$$

Integrality yields $G \leq f$. Hence $f = G = n!$.

Constructive completeness. Set $f = G = n!$. Equation (A18) gives $G - \frac{1}{2} < V/D \leq G$, so its ceiling is G. Explicitly define

$$u = V + D - DG - 1, \qquad v = DG - V.$$

The ratio inequalities show that these are natural integers, and direct substitution verifies both equations in (A15). The proof includes $n = 1$. □

The output f occurs in its own radius, but no numerical fixed-point assumption is made. It is an existential arithmetic coordinate. The equations themselves force its unique intended value.

3.5 The digit graph and a noncircular radix bootstrap

For clarity, first state the digit interface. Suppose $q \geq 2$ and all binomial coefficients $\binom{n}{j}$, $0 \leq j \leq k$, are less than q. If

$$(q+1)^n = q^k(qh+c)+r, \qquad 0 \leq r < q^k, \quad 0 \leq c < q, \tag{A19}$$

then $c = \binom{n}{k}$. This follows by expanding $(q+1)^n$: the lower coefficients form the carry-free remainder, the coefficient at q^k is c, and every higher term is divisible by q^{k+1}. Conversely the true coefficient supplies the ordinary quotient and remainder in (A19).

If $N \mid q$, the fused graph

$$(q+1)^n = q^k Nh + r, \qquad 0 \leq r < q^k \tag{A20}$$

is equivalent to $N \mid \binom{n}{k}$, under the same lower-digit bounds. Indeed the quotient by q^k is $\binom{n}{k} + qH$. It is divisible by N precisely when the coefficient is. All reverse witnesses are nonnegative ordinary quotients and remainders. The inequalities in (A19)–(A20) are implemented by natural slack rows.

In the final construction $t \geq 16$, and the proposed factorial output is denoted F. Define, by ordinary gates and power requests,

$$\rho = Ft^2, \quad V = (\rho+1)^t, \quad T = F^2, \quad B = \binom{T+t}{t}, \quad R = 2+t+x+y+F, \quad K = 2R, \quad N = B^K,$$

and use one common radix

$$q = N(N+V+1)^K. \tag{A21}$$

At this point the binomial symbols for the proposed D, B are *requests*; their intended values must still be justified. Impose (A15) for $n = t, f = F$, the raw index row

$$C + T + 1 = Bq_I, \tag{A22}$$

and the exact/fused digit rows. The index C is positive.

The soundness order is as follows.

1. The raw equation (A22), with $T = F^2 \geq 0$, forces $B > 0$. The base $\rho + 1$ is positive regardless of the proposed F. The independently proved positive-power graph therefore recovers $V \geq 1$.

2. The raw ceiling rows give $D = u + v + 1 \geq 1$ and $V + v = DF$, hence $F \geq 1$, without yet interpreting D. Also $K > 0$. Recovering $N = B^K$ gives $N \geq 1$, and recovering the power in (A21) gives $q > V^K \geq V^2$.

3. Because $\rho = Ft^2 \geq t$, for $0 \leq j \leq t$,

$$\binom{\rho+t}{j} \leq \binom{\rho+t}{t} \leq (\rho+1)^t = V < q.$$

   The first inequality is monotonicity up to the midpoint; the second follows termwise from $\binom{\rho+t}{t} = \prod_{i=1}^{t}(1+\rho/i)$. Thus (A19) validates D. The factorial lemma now gives $F = t!$.

4. Independently of any intended value of B,

$$F^2 + t \leq (Ft^2+1)^2 = (\rho+1)^2.$$

   Hence every coefficient through index t of its binomial request is at most $(F^2+t)^t \leq V^2 < q$. The second exact digit is valid, so $B = \binom{F^2+t}{t}$. Only now use the reciprocal-modulus lemma to conclude $B \geq 2$, and therefore $N \geq 2^K$.

All remaining power bases are $q, q+1$ or already recovered positive quantities. The simultaneous Pell criterion applies to the finite power family with its original natural witness conditions. The recovery does not assume a binomial value before that binomial's digit has been validated.

For the nine independently chosen representatives (eight columns and the index), retain the aggregate bound

$$C + \sum_{j=1}^{8} A_j + g = 9N^2. \tag{A23}$$

Every ordinary range lower index satisfies $1 \le k \le R-1$. For $0 \le j \le k$, using $K = 2R$,

$$\binom{n}{j} \le n^k \le 9^k N^{2k} < 2^{4R} N^{2k} \le N^{2k+2} \le N^K < q. \tag{A24}$$

If $j > n$, the coefficient is zero. This validates every ordinary range graph by (A20). Notice that no interpretation of those ranges was used to prove their digit bound.

3.6 Why the amplification coefficient can be small

The previous amplified route used $K = 8\,388\,608R$. Here $K = 2R$ suffices because the already required factorial provides a fixed large floor. Explicitly,

$$16! = 20\,922\,789\,888\,000 > 2^{44}.$$

For $E = 8\,388\,608$,

$$44E < 2^{44}, \qquad 2E \le 2^{44}. \tag{A25}$$

At $r_0 = 2^{44}$, one has $r_0^E = 2^{44E} < 2^{r_0}$. For any integer $r \ge r_0$,

$$\left(1 + \frac{1}{r}\right)^E \le \sum_{j=0}^{E} \left(\frac{E}{r}\right)^j < 2,$$

since $E/r \le 1/2$ and the geometric sum is finite. If $r^E < 2^r$, multiplication by this inequality gives $(r+1)^E < 2^{r+1}$. Integer induction proves

$$r^E < 2^r \quad (r \ge 2^{44}).$$

After the factorial bootstrap, $R \ge F = t! \ge 16!$, so

$$L(i, \mathbf{w}), J(\mathbf{w}) < R^E < 2^R < 2^K. \tag{A26}$$

This is exactly the unbounded evaluation bound needed for CRT decoding. It is used only *after* the factorial has been recovered, not as a circular premise for the earlier digit interpretation.

The preserved hybrid comparison admits another premise. If an existing carrier gives $R \ge 2^\lambda$, $\lambda \ge 8$, $E \le 2\lambda$, then $2^\lambda > 2\lambda^2 \ge \lambda E$ and $2^\lambda \ge 2E$. The same base case and induction prove $R^E < 2^R$. The final shared-factorial endpoint does not retain that old carrier; it uses the explicit factorial floor (A25).

3.7 Complete factorial-weight Horn-flow equivalence

Weight separation. For $1 \le j \le t$, put $W_j = 1 + jF$, $F = t!$. These weights are pairwise coprime. If a prime p divided W_j, W_k, it would divide $(j-k)F$ but not F. Thus $p \mid j-k$, so $p \le t$ and $p \mid F$, a contradiction. Every positive-label weight is greater than one and has a prime divisor. The truth weight is $W_0 = 1$.

Soundness of the encoded path. Suppose each row $i < t$ has a bounded zero of (A2), with $J(a, b, c, d) \leq t$. The beta equalities give

$$x = S_i + (1 + (i + 1)y)u_i, \quad x = S'_i + (1 + (i + 2)y)v_i.$$

As $S_i, S'_i \leq y$, uniqueness of remainders modulo the common overlap modulus shows $S'_i = S_{i+1}$ for $i < t - 1$. The outer endpoint rows are

$$x = F + 1 + (y + 1)u_0, \qquad y = F + 1 + r_0, \qquad x = 1 + (1 + (t + 1)y)u_t. \tag{A27}$$

They force $S_0 = F + 1 = W_1$ and $S'_{t-1} = 1$, because these also lie in the prescribed remainder intervals.

Every summand in (A2) is nonnegative. Thus its Horn product vanishes, and at least one actual rule branch vanishes. Choose such a branch and denote its premises by p_i, q_i and conclusion by v_i, using truth label 0 for unused premises. The branch equation is

$$S'_i W_{v_i} = S_i W_{p_i} W_{q_i}. \tag{A28}$$

For an axiom both premises are 0; for generalization one is 0. The truth branch has all three labels 0. The bundled schema factor is a sum of two squares, so its vanishing also enforces its actual side condition. A purely numerical match with an invalid schema instance is not accepted.

All labels in (A28) are in $[0, t]$, by the definition of J. All weights are positive. Starting from $S_0 = W_1 > 0$, equation (A28) inductively forces every S'_i, hence every S_i, positive. Multiplying the row equalities and cancelling the overlapping positive states is therefore legitimate. The endpoints yield

$$\prod_i W_{v_i} = W_1 \prod_i W_{p_i} W_{q_i}. \tag{A29}$$

For any positive label $j \leq t$, choose a prime $p_j \mid W_j$. Pairwise coprimality annihilates the valuations of every other weight. Taking the p_j-valuation in (A29) and cancelling the positive integer $v_{p_j}(W_j)$ gives

$$\#\{i : v_i = j\} - \#\{i : p_i = j\} - \#\{i : q_i = j\} = \delta_{j1}. \tag{A30}$$

The weights need not be prime or square-free.

Let D_0 be the closure of truth under the finitely many chosen valid rules. If falsity 1 were outside D_0, each rule with conclusion outside D_0 would have at least one premise outside D_0. Hence the total number of outside conclusions would be at most the number of outside premises. Summing (A30) over the positive labels outside D_0, however, gives the difference equal to 1. This contradiction proves $1 \in D_0$, hence a finite actual refutation. This argument admits cyclic, repeated and unordered rule occurrences. No fictitious balance equation for label 0 is used: truth is already in D_0.

Constructive completeness. Start from a finite C222 derivation of falsity and expand its proof DAG to a finite proof tree. Collect its finitely many parameter tuples and compute the full value of J for each tuple, including the codes of branches not used by that tuple. Also include the fixed truth-padding tuple $a = b = c = d = 0$. First choose t at least 16, at least the number of tree steps, at least every parameter, and at least every one of those finitely many label sums. This choice does not involve F or any accumulator. Only then define $F = t!$ and the weights.

Start with the pending multiset containing falsity. Expand a proof-tree conclusion v backwards to its premises p, q. The product of weights of the pending multiset before the expansion is S; the product after it is S'. Both are positive integers and satisfy (A28). Truth leaves have unit weight. Once the proof tree is exhausted the product is 1. Insert truth steps to obtain exactly t rows; they leave the current product unchanged and their label sums were included in the horizon choice.

Now choose y divisible by $t!$, at least every accumulator, and at least $F + 1$. The $t + 1$ moduli $1 + (i + 1)y$, $0 \leq i \leq t$, are pairwise coprime: a common prime would divide an index difference at

most t while remaining coprime to y, impossible because $t! \mid y$. CRT gives a nonnegative x with all prescribed state residues. Every residue lies in $[0, y]$, below its modulus, so the corresponding Euclidean quotients are natural and at most x. The outer slacks and quotients in (A27) are natural as well. This constructs the bounded eight-column source path. In particular, the horizon is not required to dominate the accumulator sizes, which are chosen and encoded only afterwards. □

3.8 Derived polynomial ranges and their independent carry bound

General derived-range lemma. Use the amplified CRT construction above and now require $K \geq t + 1$. Let J be an integer polynomial nonnegative on natural inputs, with $0 \leq J(\mathbf{w}) < 2^K$ for every bounded local tuple. Adding

$$N \mid \binom{J(\mathbf{A})}{t+1} \tag{A31}$$

forces $J(\mathbf{w}) \leq t$ in every decoded row.

Indeed, for the relevant prime power $p^e \mid N$, (A8) gives $J(\mathbf{A}) \equiv r \pmod{p^e}$ for some $0 \leq r \leq t$. Polynomial congruence gives $J(\mathbf{w}) \equiv r \pmod{p^e}$. Because $J(\mathbf{w}) < 2^K \leq p^e$ and $t < K \leq p^e$, these are both least nonnegative residues. Thus $J(\mathbf{w}) = r \leq t$.

Conversely, suppose the selected local tuples have $J(\mathbf{w}) \leq t$. The precision-$2K$ representatives give $J(\mathbf{A}) \equiv J(\mathbf{w}) \pmod{p^{2Km}}$. If $J(\mathbf{A}) < t + 1$, its binomial is zero. Otherwise its falling factorial contains a factor divisible by p^{2Km}. Division by $(t + 1)!$ loses at most $t < K \leq Km$ in valuation, so (A31) follows for every prime of B. This is the complete reverse direction and does not require any independent CRT coordinate for J. □

For the specific label expression (A3), the needed local bound is (A26). Thus (A31) replaces the extra witness w and residual $(t - w - J)^2$ of the nine-column predecessor. The local source (A2) itself is unchanged.

There is an important distinct implementation obligation: the digit graph for (A31) must be carry-free even though its upper argument is a derived polynomial. The aggregate (A23) must *not* include $J(\mathbf{A})$, which may be much larger than N^2. The ordinary CRT construction cannot promise such an aggregate bound for a derived value.

Let n be the largest of the four encoded constructor parameters. By (A23), $n \leq 9N^2$; because $N \geq 2^K$ and $t \geq 16$, $N \geq 10$. Hence

$$n + 1 \leq 9N^2 + 1 \leq N^3.$$

The independently certified positive-DAG bound (A5) gives

$$J(\mathbf{A}) \leq (n + 1)^\ell \leq N^{3\ell}. \tag{A32}$$

The finite numerical base check is

$$2 \cdot 16! \geq 3\ell \cdot 17.$$

It persists for every $t \geq 16$: if $2t! \geq 3\ell(t + 1)$, then

$$2(t + 1)! \geq 3\ell(t + 1)^2 \geq 3\ell(t + 2).$$

Consequently

$$3\ell(t + 1) \leq 2F < 2R = K. \tag{A33}$$

For every lower digit $0 \leq j \leq t + 1$, (A32)–(A33) yield

$$\binom{J(\mathbf{A})}{j} \leq J(\mathbf{A})^{t+1} \leq N^{3\ell(t+1)} \leq N^K < q. \tag{A34}$$

Again a coefficient above the upper index is zero. This proves that the additional fused digit graph recognizes exactly (A31). Its bound is independent of the representative aggregate and explicitly accounts for the entire constructor degree and coefficients. There are ten range requests in total, but only nine independent representatives in (A23).

3.9 Assembly into one complete polynomial

The finite arithmetic system now consists of the outer and endpoint rows, the single self-bounding factorial, the reciprocal product binomial, the amplification and common radix, nine ordinary ranges, one derived-label range, the index congruence and the local evaluation divisibility. Every power request is replaced by the inherited explicit simultaneous Pell graph. Every ordinary gate is quadraticized while sharing identical intermediate expressions. If the resulting quadratic residuals are $r_1, \dots, r_s$, define

$$Q(\mathbf{z}) = \sum_{\nu=1}^{s} r_\nu(\mathbf{z})^2. \tag{A35}$$

Its complete literal rows and expanded coefficients are supplied. For natural inputs, $Q = 0$ is equivalent to every row being zero.

Soundness of the assembled equation. From a natural zero, recover the true powers and interpret the exact factorial and reciprocal-product digits in the order proved above. Then use the factorial floor to establish (A26), and the separate ordinary and derived carry estimates to interpret all range graphs. Amplified CRT decoding yields a bounded zero of (A2) at every row, and the derived-range lemma gives $J \le t$ there. The factorial-weight Horn-flow theorem produces a finite C222 refutation. Applying the inherited source calibration gives the stated ZFC refutability predicate.

Constructive completeness of the assembled equation. Start with a finite calibrated refutation. Choose the padded horizon and construct the shared-factorial path before choosing its beta encoding, as proved above. Set $F = t!$, $T = F^2$, $B = \binom{T+t}{t}$, $K = 2R$, and $N = B^K$. Choose the eight columns and the index by precision-$2K$ CRT. Their positive representatives are at most N^2, so (A23) has a natural slack. The local polynomial remains nonnegative on all natural inputs; its divisibility quotient is therefore natural. The derived-label lemma supplies (A31). Define the true ρ, V, q; the factorial lemma supplies its two ceiling witnesses, and the carry-free estimates supply the actual digit quotients and remainders. Finally the existing simultaneous Pell completeness theorem supplies one common frame and all witnesses for this finite family of true positive powers. Thus the full quadratic system, and hence (A35), has a natural zero.

The final nonnegative contraction deletes coordinates only with exact integral reverse substitutions certified to be nonnegative on the remaining natural variables. For signed-coefficient restorations, the certificate explicitly expands a positive rational sum of squares of affine forms plus a coefficientwise nonnegative remainder. Every identity and substitution is independently replayed, so the contraction preserves both zero-existence directions, not merely the forward map. It leaves 349 variables and 268 quadratic rows in the Cantor endpoint. Independent expansion of their squares gives the stated exact quartic and coefficient ledger.

Measured tradeoffs and finite evidence. The new endpoint uses one factorial, two exact binomials, ten range graphs, nineteen positive-power requests, sixteen independently tested Pell indices and one derived index. Normal and Python-optimized regeneration reproduce all twelve final endpoint files byte for byte. The semantic regressions include 361 200 range cases, 32 constructive CRT cases, ten derived-range constructions, 600 factorial candidates and 47 904 small Horn-rule multisets. Negative controls expose the false converse to (A8), invalid canonical index lifting, a bad CRT row, missing label bounds, and derived values that exceed the representative aggregate. These finite tests support the implementation; the quantified proofs above establish the unbounded implications.

The 350-variable local-slack predecessor has fewer monomials than the 349-variable derived-range endpoint. The earlier 363-variable reciprocal construction has fewer still. These are explicit variable/support/height tradeoffs; no global optimum or worldwide priority is asserted. All

inherited code, calculus, Pell and contraction dependencies retain their original provenance and notices.

4 A signed direct certificate in 519 integer variables

The principal unrestricted-integer endpoint is

$$P_{519}(z_1, \dots, z_{519}) = \sum_{j=1}^{321} R_j(z)^2,$$

where every R_j is the literal integral quadratic listed in the accompanying equation file. Its exact degree is four, its collected expansion has 6253 nonzero monomials, and its largest absolute coefficient is 411060. There are no implicit sign conditions, variable exponents, unspecified arithmetic ports, or program parameters in this polynomial. Its ordinary mathematical correctness proof gives

$$\mathrm{ZFC} \vdash \left[\forall z \in \mathbb{Z}^{519}\ P_{519}(z) \neq 0\right] \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

This is an equivalence, not a proof of consistency. No new complete Lean proof or object-language derivation is asserted.

4.1 The signed arithmetic interface

Start with the square-shell reciprocal prime-power source in the preceding sections, retaining the large carrier

$$M = M_0 + 1 = ((t+1)^t)^{2^{22}}, \qquad R = 2 + t + x + y + M_0, \qquad K = 2R.$$

Let F denote the proposed factorial, $\rho = Ft^2$, $V = (\rho+1)^t$, $B = \binom{F^2+t}{t}$, $N = B^K$, and $q = N(N+V+1)^K$. Until the arithmetic graphs are justified these are names for their proposed outputs, not assumptions about their meanings.

Replace the common Pell budget by

$$W = 2 + \sum_{a \in \mathcal{A}} a^2,$$

where $\mathcal{A}$ is the finite set of distinct actual base, exponent, and output atoms of the nineteen power requests. The atom list and its affine expansion certificate are supplied. Thus $W \geq 2$ and $|a| < W$ without a sign premise on any power output. Retain the shared norm

$$A^2 - W^3(W+2)(z+1)^2 = 1, \qquad A, z \geq 0.$$

Pell classification at base $W+1$, together with $Y_j(W+1) \equiv j \pmod{W}$, gives $A \geq (W+1)^W$. Therefore, for $1 \leq n < W$,

$$T_n = 2An - n^2 - 1 > W^W + W > 0.$$

The finite indexed-Pell certificate inherited from the prime-power construction admits signed comparison coordinates and signed principal abscissa. Its hypotheses are $A \geq 2$, comparison base at least two, $|U| \geq 2$, $1 \leq k \leq Y$, the three norm equations, the divisibility $Y^2 \mid V_{\mathrm{aux}}$, and its four congruences. It gives

$$|X| = X_k(A), \qquad Y = Y_k(A).$$

For completeness, the signed extension follows by classifying absolute values of all three norm solutions. The abscissa congruence gives the comparison index congruent to either sign of the principal index modulo $2n$, and the ordinate congruence gives either sign of k modulo $4Y$. Squared

divisibility gives $Y \mid n$. Consequently the principal index is congruent to k or $-k$ modulo $2Y$. Both lie in $[1, Y]$, so they coincide, including the endpoint case. This is the same indexed-Pell lemma used by the inherited 553-integer proof, not a finite-test inference.

For the signed power row

$$X + (n - A)Y = m + T_n u$$

with $1 \le n, k < W$ and $|m| < W$, a positive X yields $m \equiv n^k \pmod{T_n}$ and hence $m = n^k$ by the strict interval bound. A negative X instead gives $mn^k \equiv -1 \pmod{T_n}$: multiply the two conjugate expressions and use the norm. Since $|mn^k + 1| < T_n$, this forces $mn^k = -1$. It is impossible for $n \ge 2$; for $n = 1$ it allows only $m = -1$. Thus a single request with base at least two orients the shared index pair for all requests at that exponent.

4.2 Why the auxiliary nonzero guard disappears

The earlier 553-integer construction inserted a three-witness guard for $U^2 \ge 2$. It is unnecessary in the present reciprocal source. Every seed exponent is already positive before any power has been interpreted. The complete list consists of

$$t,\ K,\ Ft^2 + t,\ F^2 + t,\ y + 1,\ x + 1, \quad \text{and the positive encoded representatives.}$$

The seed $t+1$ is derived by the existing addition identity and is positive as well. The retained seed guards imply $t \ge 1$, $x, F, r_0 \ge 0$ and positive encoded representatives. The final carrier squaring already implies $M_0 + 1 \ge 0$, whether or not its power input has the intended value. Hence

$$y = M_0 + r_0 + 2 \ge 1, \qquad R = t + x + r_0 + 2(M_0 + 1) + 2 \ge 3.$$

Every seed inequality $Y_i = k_i + e_i$, $e_i \ge 0$, therefore gives $Y_i \ge 1$. The unchanged common auxiliary product

$$V_{\text{aux}} = \prod_i Y_i^2\,(v + 1), \qquad v \ge 0,$$

is positive. Its norm yields $U^2 \ge A^2 \ge 4$, proving the omitted guard without a circular use of any power output. The emitted certificate expands each seed exponent after substituting $M_0 = Z - 1$, $y = Z + r_0 + 1$ with $Z \ge 0$; every resulting polynomial has nonnegative coefficients and positive constant term.

4.3 The noncircular recovery order

Sixty natural seed coordinates are guarded by $s = a^2 + b^2 + c(c + 1)$. Legendre's three-square theorem applied to $4s + 1$ proves that this represents exactly the nonnegative integers. The coordinate list and an ordered exact sign-inference certificate are part of the artifact; all other coordinates are unrestricted.

Let D be the factorial binomial output, and let r, s, o be its retained nonnegative remainders and slacks. Before interpreting any binomial, the actual raw rows imply

$$D = r + s + 1 \ge 1, \qquad q = D + o + 1 \ge 2.$$

The existing request q^t therefore orients the t-indexed pair. The request $(t + 1)^t$ now has its true value, recovering $M_0 \ge 0$ and the usual endpoint signs. The request $(Ft^2 + 1)^t$ is at a positive base, possibly initially one, and shares the already oriented pair; hence $V \ge 1$. The raw factorial rows give $DF = V + s$, so $F \ge 1$ and then $V \ge 2$.

The reciprocal index row is

$$C + F^2 + 1 = Bi_q.$$

Here $C \geq 1$ and the output B is retained nonnegative, so $B > 0$. The request B^K either has its intended positive output N, or its only remaining inverse case has $B = 1$ and $N = -1$. In both cases the second K-indexed base satisfies

$$N + V + 1 \geq 2.$$

That request orients the K pair and excludes the inverse case. Thus $N = B^K > 0$ and the radix has its intended value. Every remaining seed has an actual request at base q or $q + 1$, both at least two. All its exponents are positive by the preceding subsection. This recovers all nineteen powers, without first assuming the factorial, product-binomial, or range-binomial conclusions under validation.

The natural source's ordered factorial, low-digit, prime-power lifting, and CRT arguments now apply unchanged. Required Euclidean quotients are nonnegative automatically from their positive divisors and bounded nonnegative remainders. The auxiliary congruence quotients and comparison coordinates need not become nonnegative; only the proved Pell congruences use them. Simultaneous Pell completeness supplies compatible witnesses for any genuine semantic certificate.

4.4 Factoring the Horn squares over the integers

Write the nine ordinary axiom balances, including truth, and the two inference balances as $L_1, \ldots, L_{11}$. Let D_b be the guarded bundle balance and

$$G = (c - a)^2(c - b)^2 - (d + 1)^2.$$

The original Horn summand is

$$\left(\prod_{i=1}^{11} L_i^2\right)(D_b^2 + G^2).$$

Instead form signed intermediate products and one final square:

$$H = \prod_{i=1}^{11} L_i, \qquad H^2(D_b^2 + G^2).$$

These are identically equal over every commutative ring. Likewise form $h = (c - a)(c - b)$ and $G = h^2 - (d+1)^2$ before squaring G. This removes duplicated square gates. The two beta squares remain outside the Horn product. Merely replacing H^2 by H in their common sum would be unsound because of cancellation; that change is not made.

The factorization certificate identifies every original square root, the eleven product links, and every changed row. It checks that no intermediate factor escapes this subnetwork. Given an old zero, define the new signed factors and products; given a new zero, restore the old squared factors and their products. The visible Horn value and all other coordinates agree exactly. Existing sign guards mention none of the changed internal coordinates. Therefore the change preserves integer-zero existence in both directions, independently of arithmetic or logical assumptions.

4.5 Projection and integral translation

After the sixty sign guards, the signed source has 734 coordinates and 536 rows. Factorization has 735 coordinates and 535 rows before pruning. The recorded monic projection deletes 214 coordinates, of which 168 have affine and 46 have quadratic integral restorations, and removes two unconstrained coordinates. It leaves 519 coordinates and 321 rows. Every deleted defining row is exactly $z - Q$ up to sign. A quadratic Q is substituted only where z occurs linearly in every other row, so all residuals remain quadratic. Reversing the substitutions gives the unique integral lift of every deleted defined coordinate.

Finally translate each retained encoded seed, t_0, M_0, and retained coordinate whose name ends in `minus1` by minus one: the convention is $z_{\text{old}} = z_{\text{new}} - 1$. The exact coordinate indices, not

the names, define the transformation in its certificate. Its inverse adds one, so this is a bijection of $\mathbb{Z}^{519}$. It absorbs offsets that otherwise inflate constant and linear coefficients. The degree and variable count stay unchanged, while monomials fall from 6816 to 6253 and coefficient height from 6474801 to 411060. This is a coordinate choice, not a claim of globally minimal height.

4.6 Finite evidence and logical scope

The independent-friendly artifact consists of the raw quadratic source, square-budget and sign certificates, source-factorization certificate, monic elimination certificate, integral translation certificate, final quadratic residual list, and fully collected quartic coefficients. The standard-library reproduction script imports no downloaded builder. Exact coefficient reconstruction checks these finite algebraic claims; the unbounded Pell, CRT, and proof-flow arguments establish their meaning.

The numerical-calculus calibration remains `KEndpoint.c222_iff_zf_refutation` followed by `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. Square-shell recursive recoding fixes truth and falsity and transports all formula and variable positions. No K215 execution, Jones polynomial, or Sun variable reduction occurs in this construction. Full transitive formal-kernel checking, global minimality, and worldwide priority are not claimed.

4.7 The separate 517 variable tradeoff

Applying the same signed compiler to the derived-label factorial-carrier source yields a second quartic in 517 unrestricted integers, with 310 residuals, 6774 collected monomials, and coefficient height 3904267. The complete arithmetic proof for that source is separate: it replaces the large carrier by $F = t!$ and certifies the derived label sum by a range binomial rather than a ninth independent local coordinate. It is not a further projection of P_{519}.

The signed proof adapts without an additional guard. Here $y = F + r_0 + 1 \geq 1$ and $K = 2(2 + t + x + y + F) > 0$ immediately. The additional seed exponent is the derived label sum J; its constructor circuit uses only nonnegative addition and multiplication and has an explicit positive constant summand. Therefore $J \geq 1$ before power recovery. The emitted version-two positivity certificate follows these gates compositionally, retaining positive lower bounds capped at 64; it does not expand the enormous polynomial J. All other seed indices are the same positive types already listed. Thus the product again forces $V_{\mathrm{aux}} > 0$ and $|U| \geq 2$ without a guard. The raw $q \geq 2$ argument orients t directly, the factorial ceiling forces $F \geq 1$, and the same two-request argument orients K. Source factorization, integral projection, and the explicitly recorded coordinate translation then establish the 517-variable endpoint. It improves the variable count by two but has more terms and larger coefficients than the primary 519-variable presentation; neither dominates the other.

5 Comparison with the pinned frontier

The comparison is frozen on 24 October 2026 at public repository commit

`84e16e43092614bae262ca59651cb9d77e31a255`.

The active research is substantially ahead of the archival manuscript-collection README. The supplied prime-power and reciprocal-shell packets are included as baselines even where they have not yet reached that public snapshot.

All following direct polynomials are quartics. Here r counts quadratic residuals, s counts collected monomials, and h is the maximum absolute coefficient after expanding the single sum of squares.

Construction	Domain	Variables	r	s	h
Prime-power	$\mathbb{N}$	375	294	7907	h_N
Prime-power	$\mathbb{Z}$	553	348	7911	h_Z
Reciprocal, old syntax	$\mathbb{N}$	363	291	6674	83842716
Reciprocal, shell syntax	$\mathbb{N}$	364	292	6613	7002666
Signed shell	$\mathbb{Z}$	660	386	6408	3877579
Public75, default	$\mathbb{Z}$	604	372	9676	2445790974704
Public75, affine16	$\mathbb{Z}$	609	377	8823	964213
Round 14 centered	$\mathbb{Z}$	519	321	6253	411060
Round 14 derived	$\mathbb{Z}$	517	310	6774	3904267
Round 14 derived labels	$\mathbb{N}$	349	268	12768	86915503

The large baseline heights are

$$h_N = 495176015714153095122207427825 4, \qquad h_Z = 534790096971332833190424347930.$$

All seven baseline entries were independently re-expanded from JSON residuals and compared coefficient by coefficient with the delivered polynomial, without importing a producer. The three round 14 principal entries also passed independent full expansion, source-projection, and coordinate-translation audits.

5.1 What is and is not dominated

The 519-integer quartic strictly improves the supplied 553-integer and 660-integer equations, and the pinned public 609-integer equation, simultaneously in variables, collected monomials, and coefficient height, at unchanged degree. Its 411060 bound applies to every expanded coefficient, not only a radix or construction constant. The 517-integer companion saves two more unknowns but has 521 more monomials and a higher coefficient bound; neither new integer endpoint dominates the other in all three measures.

The 349-natural endpoint saves fourteen variables relative to the 363-natural reciprocal equation but has 6094 more terms and slightly larger height. Its intermediate 350-natural factorial-slack companion has 11472 terms and height 86523090, another explicit tradeoff. These domains must not be merged: nonnegative unknowns are not unrestricted integer unknowns without a further proved transformation.

The native-compiler route already retains a quartic in 37 unrestricted integer variables, with 31-variable sextic and 26-variable octic companions, at very large fixed coefficients. Thus neither 519 nor 517 is a global fewest-variable quartic record. Payload bytes, coefficient bit lengths, and the height of the expanded output polynomial are different quantities.

5.2 Small variable arithmetic circuits

Selected pinned circuit points are

domain	unknowns	operations	exact degree	
$\mathbb{Z}_{>0}$	9	405	1191570	
$\mathbb{Z}_{>0}$	10	99	368	
$\mathbb{Z}_{>0}$	10	105	340	
$\mathbb{Z}_{>0}$	11	111	211	
$\mathbb{Z}_{>0}$	12	96	70	
$\mathbb{Z}$	10	471	1245324	
$\mathbb{Z}$	12	145	1200	
$\mathbb{Z}$	12	141	1456	
$\mathbb{Z}$	13	135	536	
$\mathbb{Z}$	14	117	312	
$\mathbb{Z}$	12	141	768	(this report)

The 145- and 141-operation twelve-integer baselines and 105-operation ten-positive count were independently recomputed from live graphs; exact baseline degrees use their source proofs. The new norm lowers degree from 1456 to 768, a 47.25% reduction at the same operation count. It also reduces both degree and operations relative to the 1200-degree, 145-operation point. It does not beat every other point on every axis. No comparable fully expanded coefficient height was measured for these norms. The scalar tuple's 1064921-bit β is not interchangeable with the compact44 tuple's 590069-bit β.

5.3 Rejected shortcuts and remaining questions

The shared-projection 83-operation proposal is refuted for its original weakened interface by zeros at unbounded ordinary inputs for a fixed empty-language program. The valid universal operation baseline remains 84 operations, degree 187, and 18 positive witnesses; 92 operations, degree 325, and 17 witnesses is another tradeoff. These are universal input/witness counts, not direct fixed-ZFC certificate counts. The refutation does not refute every possible 83-operation construction, the modified75 compiler, or the unchanged source34 tuple.

Further research should target specific verifiable improvements:

1. Can a lower-degree intersective carrier satisfy the required prime-local conditions while preserving a no-rational-root bootstrap? The general saturation lemma makes the exact allowable degree and rank margin explicit.
2. Can a further integral shear reduce the degree below 768 without increasing the live operation count? Any candidate needs a polynomial identity and a nonzero specialized leading coefficient, not just a syntactic estimate.
3. Can the 517-integer branch inherit the 519-integer branch's small coefficient height? Integral shifts and monic projections should be compared using a reproducible Pareto search rather than an unqualified "simplest" score.
4. Can another positive label constructor or exact local bound improve the 349-natural support while keeping the shared-factorial proof? The derived label must remain outside the aggregate representative bound.
5. Can the signed scalar folding, full supported-modulus interpolation, and direct C222 calibration be internalized in the proof assistant? This would close a different proof obligation from finite coefficient checking.

No lower-bound theorem, global minimum, or historical priority claim is made.

6 Exact artifacts and reproducibility

6.1 The literal equation formats

The canonical norm file is `norm/equation.json`. Its `nodes` form a topologically ordered directed acyclic graph. A leaf is an integer, one of the twelve named variables, or one of six fixed coefficient ports. An arithmetic node records an operator and two earlier node indices. The output index defines the single polynomial. The adjacent coefficient binding identifies every compressed payload by raw SHA256, exact bit length, and byte length; M_F is already shifted. The companion `equation_reduced.json` is polynomial-identical.

For a quartic, the canonical `equation.json` lists variables in order and every quadratic residual. A residual term $[c, [i_1, \ldots, i_k]]$ means $c \prod_{j=1}^{k} z_{i_j}$, with zero-based variable indices, repeated indices for powers, and the empty list for the constant monomial. The equation is exactly

$$P(z) = \sum_{r \in \text{residuals}} \left(\sum_{(c,m) \in r} c \prod_{i \in m} z_i \right)^2 = 0.$$

The matching `expanded_polynomial.json` collects all equal sorted monomials and removes zero coefficients. These are finite complete definitions, not recipes with unnamed coefficients. The primary direct endpoint is `direct/integer519_centered`; the integer companion is `direct/derived16_centered_shell`; the natural endpoint is `architecture/derived16_cantor/natural`.

6.2 What the replay checks

The top-level `verify.py` uses explicit exception checks, including under Python's optimized mode. It copies the packet to a fresh temporary directory before running any builder or checker, so the delivered files are not mutated. It checks the release manifest, regenerates the four nontrivial source34 coefficient ports from the frozen machine/tile recipe, verifies all six actual payloads, rebuilds both norm graphs, verifies the generic norm identity and exact degree certificate, and independently re-expands every principal quartic.

For the direct endpoints it replays the raw arithmetic export, ordered sign closure, redundant-guard proof, exact Horn factorization, monic elimination, and integral translations. For the natural endpoint it checks every nonnegative restoration certificate, including rational weighted-square identities. It reconstructs all three principal endpoints from their bundled sources and compares generated files byte for byte. Both normal and optimized runs are included. It also rechecks seven literal baselines, three live-DAG baseline counts, modular arithmetic regressions, and deliberate negative controls.

These finite checks do not prove the unbounded Pell or logical-calibration theorems, prove consistency, or search for a full zero. The archive distinguishes producer reproduction, independent algebraic audits, mathematical review, and inherited proof-assistant obligations.

6.3 Offline commands and dependencies

The arithmetic replay requires Python 3.10 or later and SymPy for the independent resultant calculation. The remaining direct, architecture, payload, and polynomial checks use only Python's standard library. SymPy is an environment dependency, not a vendored component. Once that dependency is available, no network access or original workspace is needed. From the extracted archive root, run

```
python verify.py --report /tmp/round14-replay.json
python -O verify.py --mode optimized --report /tmp/round14-replay-O.json
```

The first command itself tests both normal and optimized child processes. The second also verifies that the top-level checks remain active under optimization. The retained legacy coefficient recipe explicitly rejects optimized mode, so the wrapper always invokes that one checked recipe normally and independently compares its output bytes. This restriction is not hidden.

Build from `report/main.tex` with pdfLaTeX; repeat until references stabilize:

```
cd report
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

TeX rendering is not part of the arithmetic proof. The final rendered PDF and its editable TeX sources are both included. Receipts give the actual command counts, file hashes, and exact byte comparisons; the manifest pins the final shipped files.

6.4 Attribution and source map

The new proof texts and code identify which transformations are new and which files were inherited. Existing licenses and attribution notices remain alongside the imported code, including ProveIt, Mathlib, and Wade-related source notices where applicable. The retained source34 machine is coefficient provenance, not an executed search for consistency. The direct branch's C222 route is separate.

Primary public references, all pinned to the reviewed commit, are:

- The active Diophantine consistency proposal review.
- The scalar-carrier12 source and its inherited proof bundle.
- The determinant and small-variable low-degree tradeoffs.
- The public direct factorial-radius sign-profile family.

The supplied Prime Power 375/553, Square Shell Reciprocal 660, and Compact Proof Certificates Round 13 packets are retained only to the extent needed for runtime reconstruction, proof boundaries, literal baseline verification, and attribution. Exploratory outputs, intake archives, upload helpers, and redundant binaries are excluded.